

Qatar-4b

R-2403

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_221226 · created 2026-08-21T00:39:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459939.6084 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.038
Transit depth (Rp/Rs)^2	1.33 +/- 0.87 [%]
Orbital Inclination (inc)	86.14 +/- 2.02
Airmass coefficient 1 (a1)	1.017 +/- 0.015
Airmass coefficient 2 (a2)	-0.013 +/- 0.037
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#1 - [350, 455]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.116 ± 0.038 vs 0.138 (deviation 0.55σ)
Duration fit vs published	0.072 d vs 0.0892 d (deviation 1.48σ)
Mid-time O-C	-8.4 min
Depth SNR	2.9
Residual scatter	1.46 %

Light curve

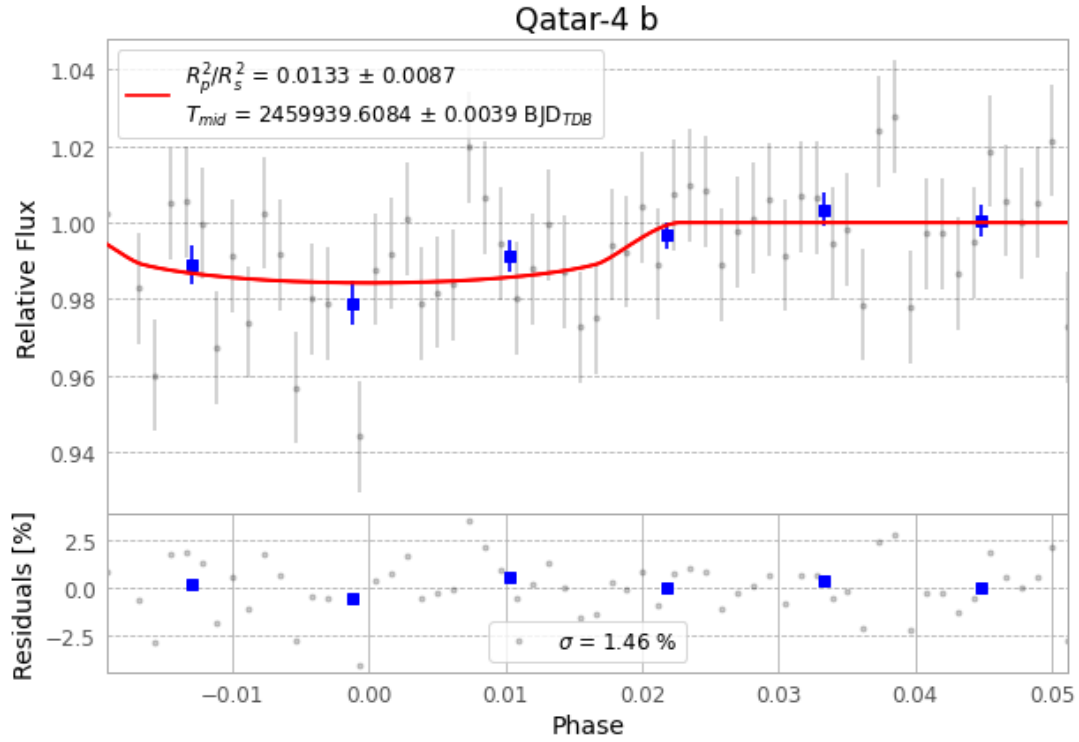


Figure 1 — FinalLightCurve_Qatar-4 b_2022-12-25.png (SHA-256 25560d4e3517b54c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [463, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f4256097965c4250...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-4221226014218.FITS → Qatar-4221226044515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-221226.log (a50a5e8d1ada...), FinalLightCurve_Qatar-4 b_2022-12-25.png (25560d4e3517...), FinalLightCurve_Qatar-4 b_2022-12-25.csv (05f7f535a551...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:39Z.